

A concept feature norm to investigate concept representations of 6–8-year-old Hungarian children

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Introduction

Concept feature norms are essential in modeling semantic organization of different populations [1]. To our knowledge, **no concept feature norm exists among 6–8 year olds**, thus investigations of idiosyncrasies in children concept networks are limited.

We created a **child-friendly website** [2] to collect a minimum of five features for concepts ($N_{\text{per-concept}} = 30$; $N_{\text{total-concepts}} = 300$) online in a population of Hungarian children aged 6–8.

Mimó offers cues if the child can not list enough, e.g., 'How can I recognize it?' or 'What was it made of?'



For a **pilot dataset**, we recruited 48 Hungarian children (23 females, $M_{\text{age}} = 7.63$, $SD = 0.94$) from a single elementary school.

We compared children data to a published dataset of young adults ('YA benchmark') [8,9] to uncover idiosyncrasies.

Methods

We allowed children to produce features on their own, then via helping cues ($M_{\text{feature number}} = 14.42$ features; $SD = 5.48$; range = 6.7–35.7). **Data collection is still on-going.**

Winged creature

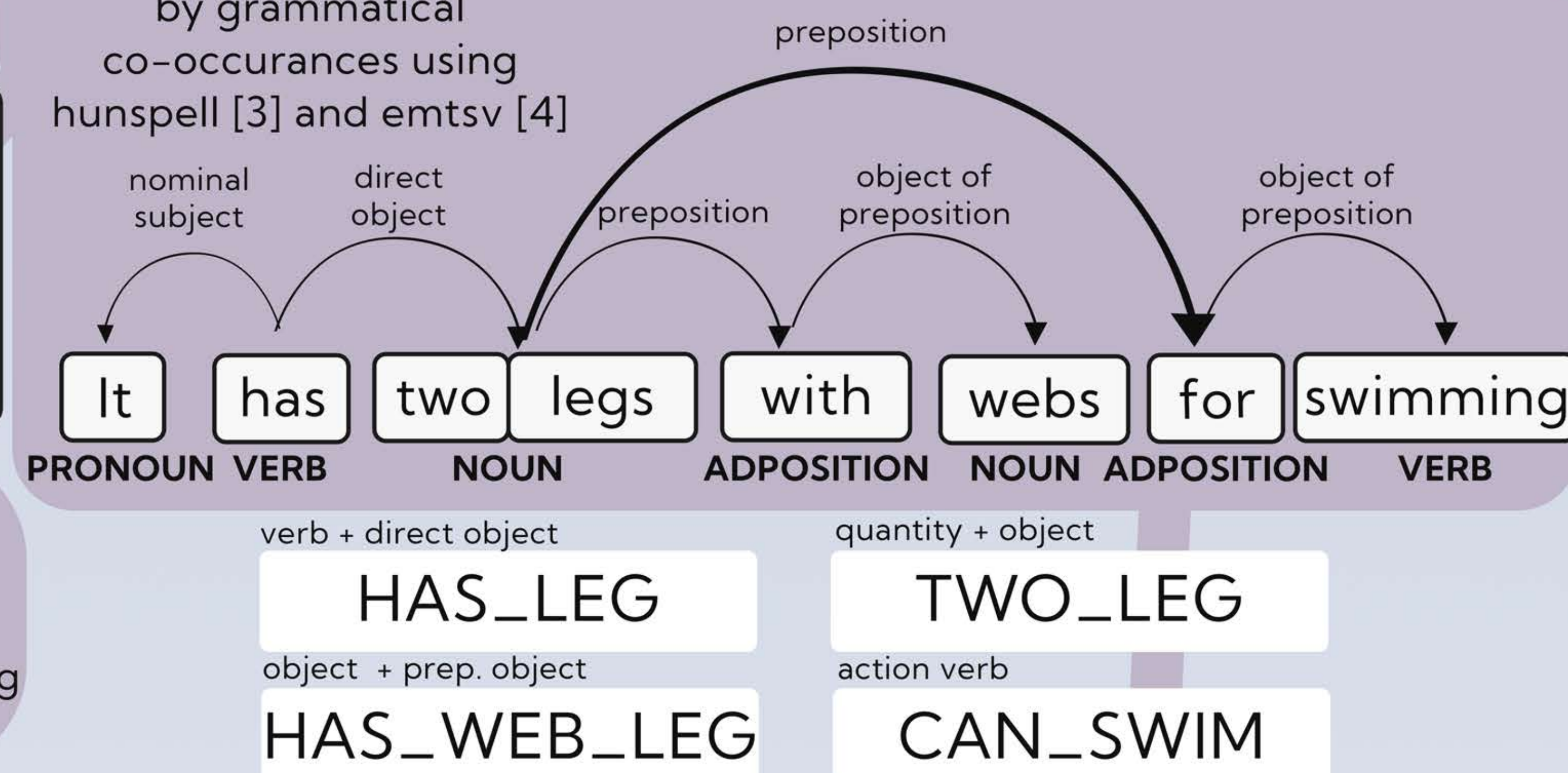
It can fly

It usually lives in lakes

It has two legs with webs for swimming

It has feathers

We extracted features by grammatical co-occurrences using hunspell [3] and emtsv [4]

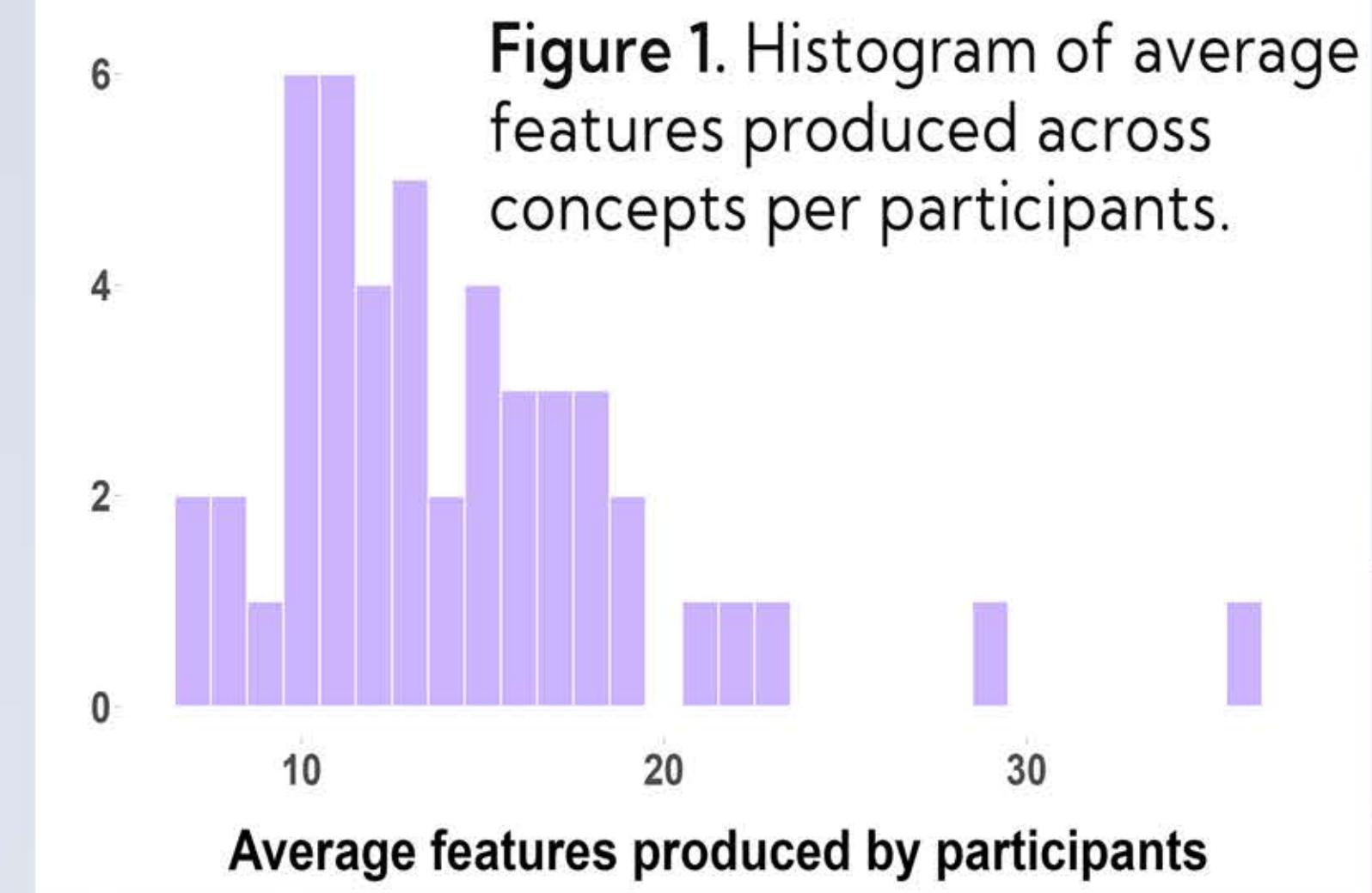
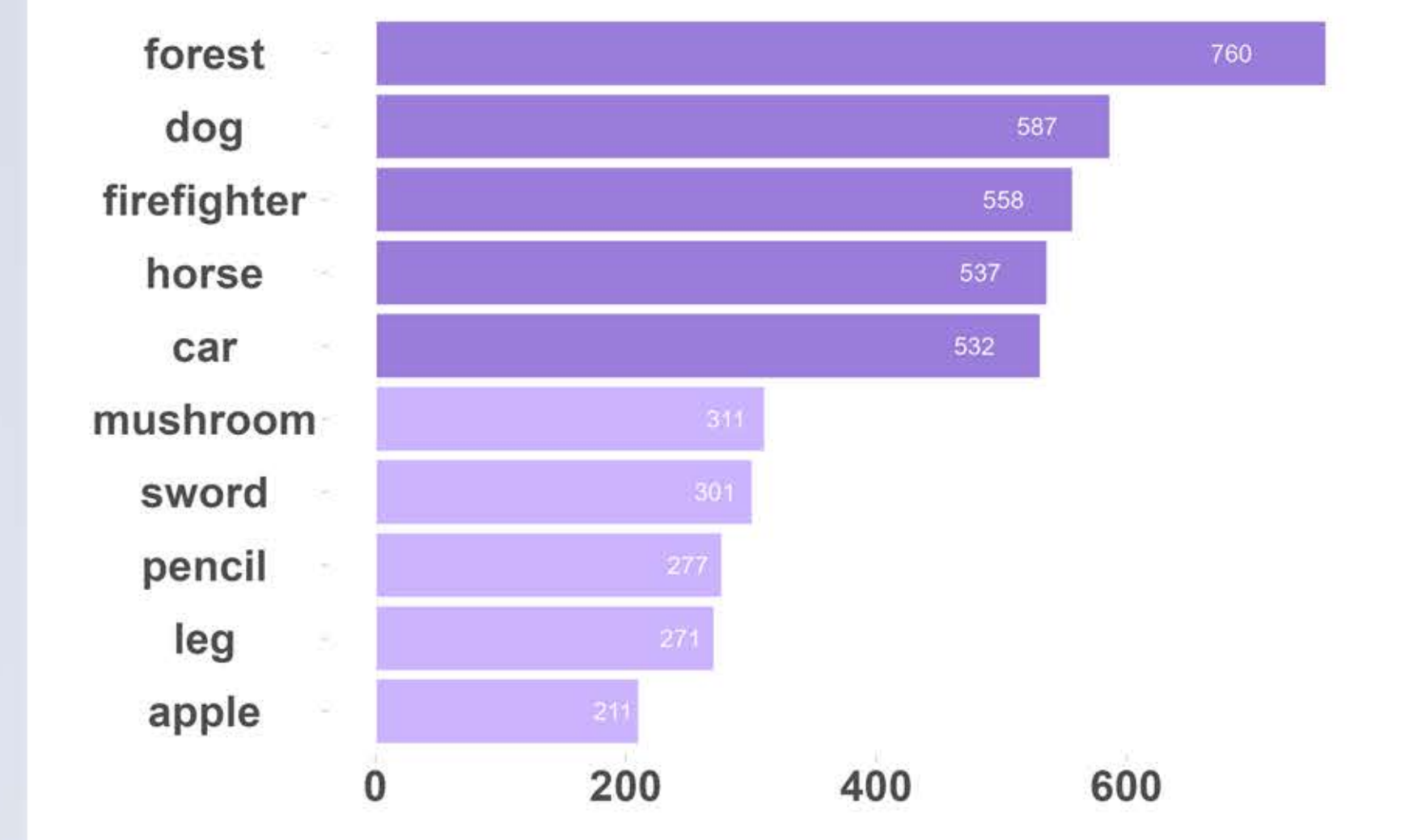


Results

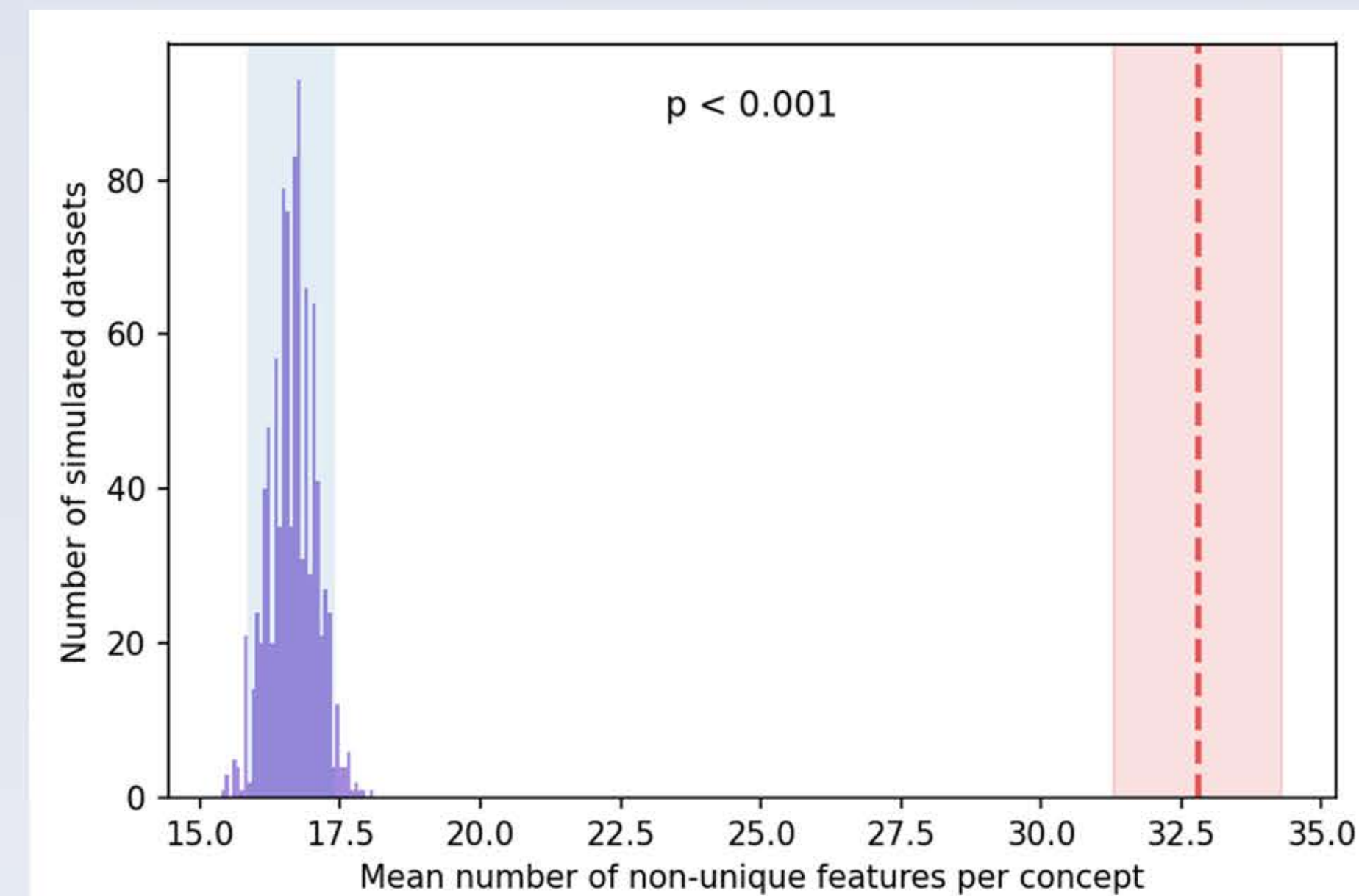
Figure 2. Bars represent the five most and five least "semantically rich" concepts, i.e., the amount of unique features collected and predicted by the following equation (x-axis):

$$S_{obs}^* + A \frac{Q_1^{2***}}{2Q_2}$$

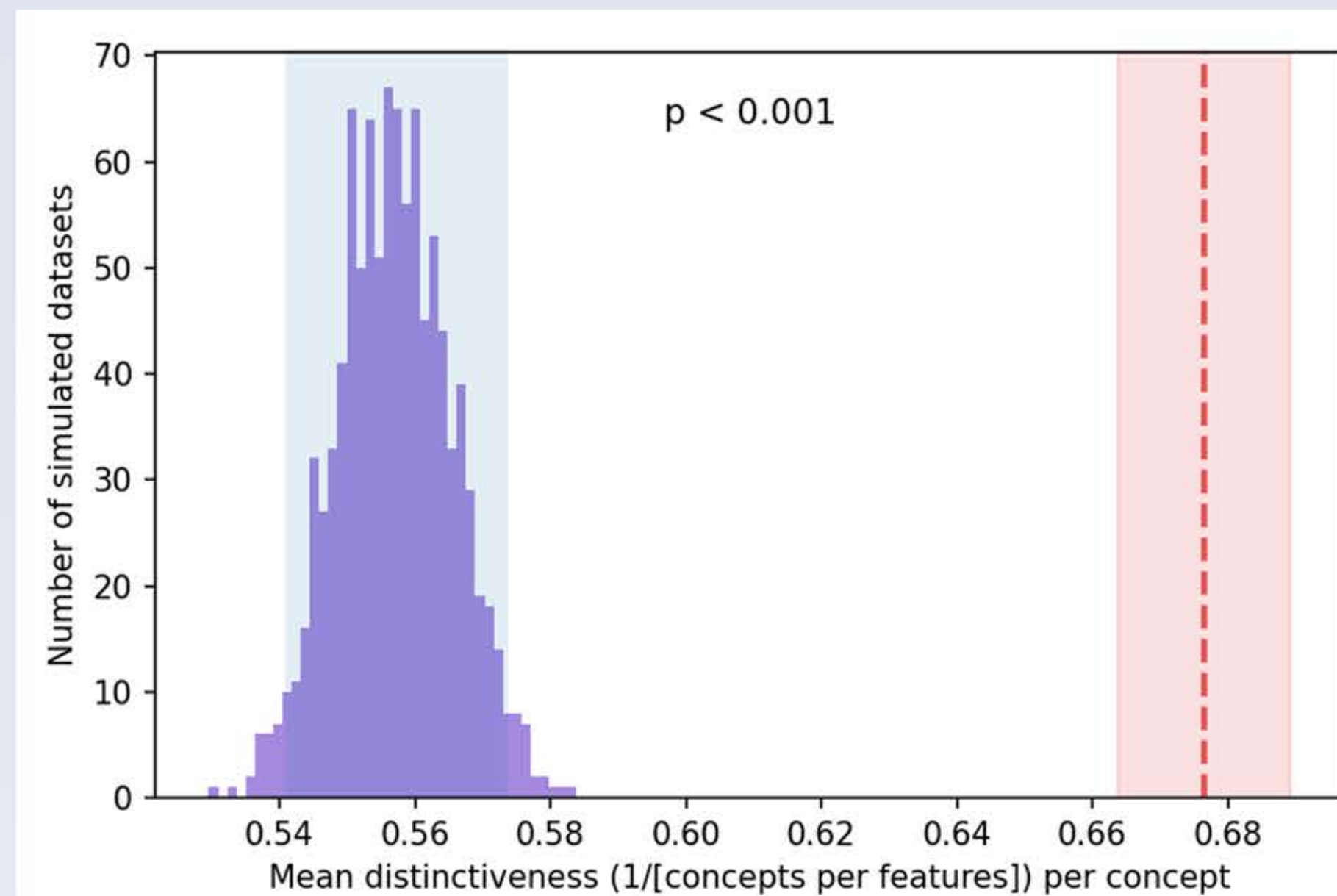
* number of features in total
** sample size-1/sample size
*** number of features mentioned by indexed number of subjects (based on [7])



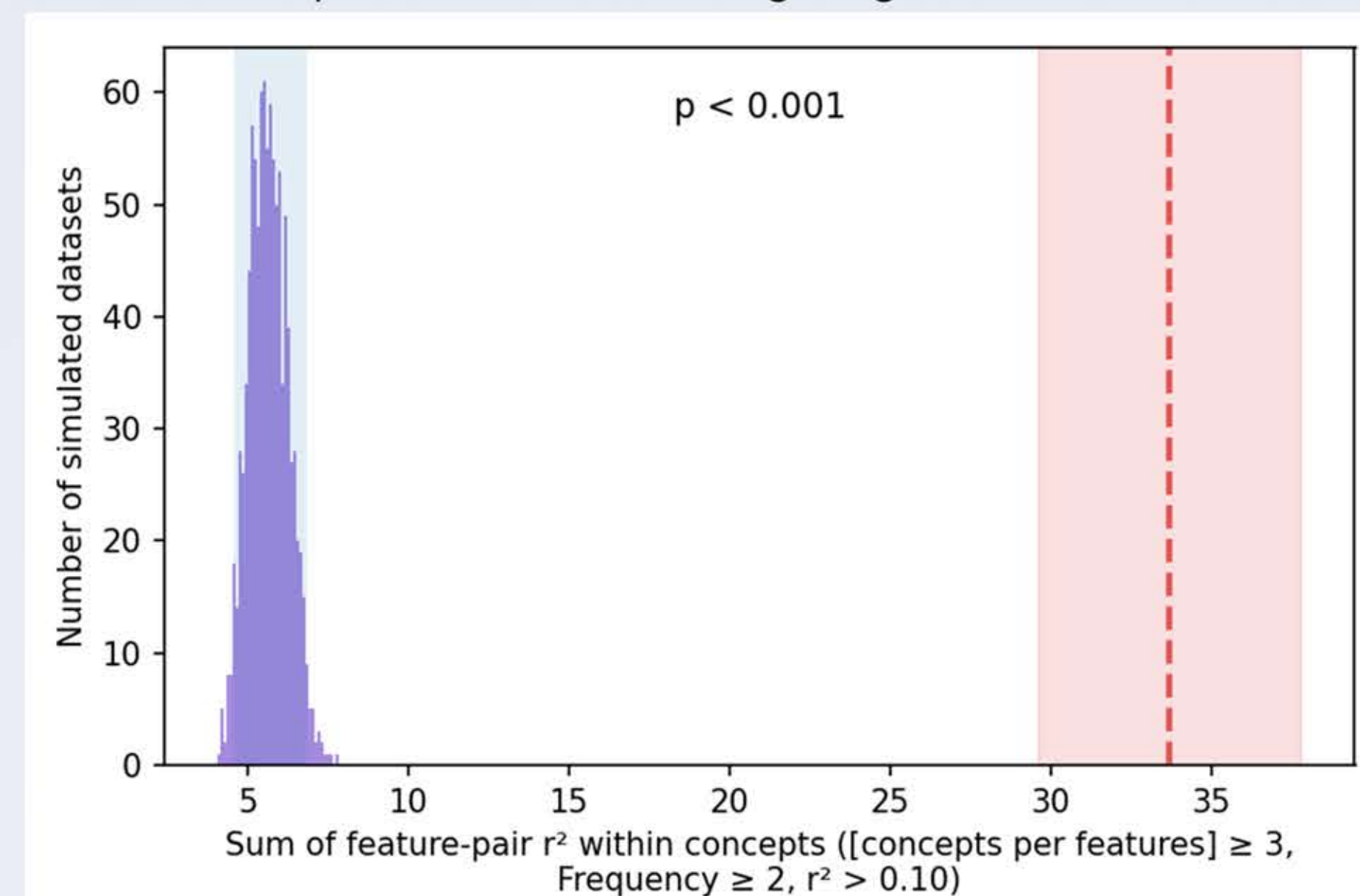
We created an N-dimensional vector space (N = number of unique features produced)



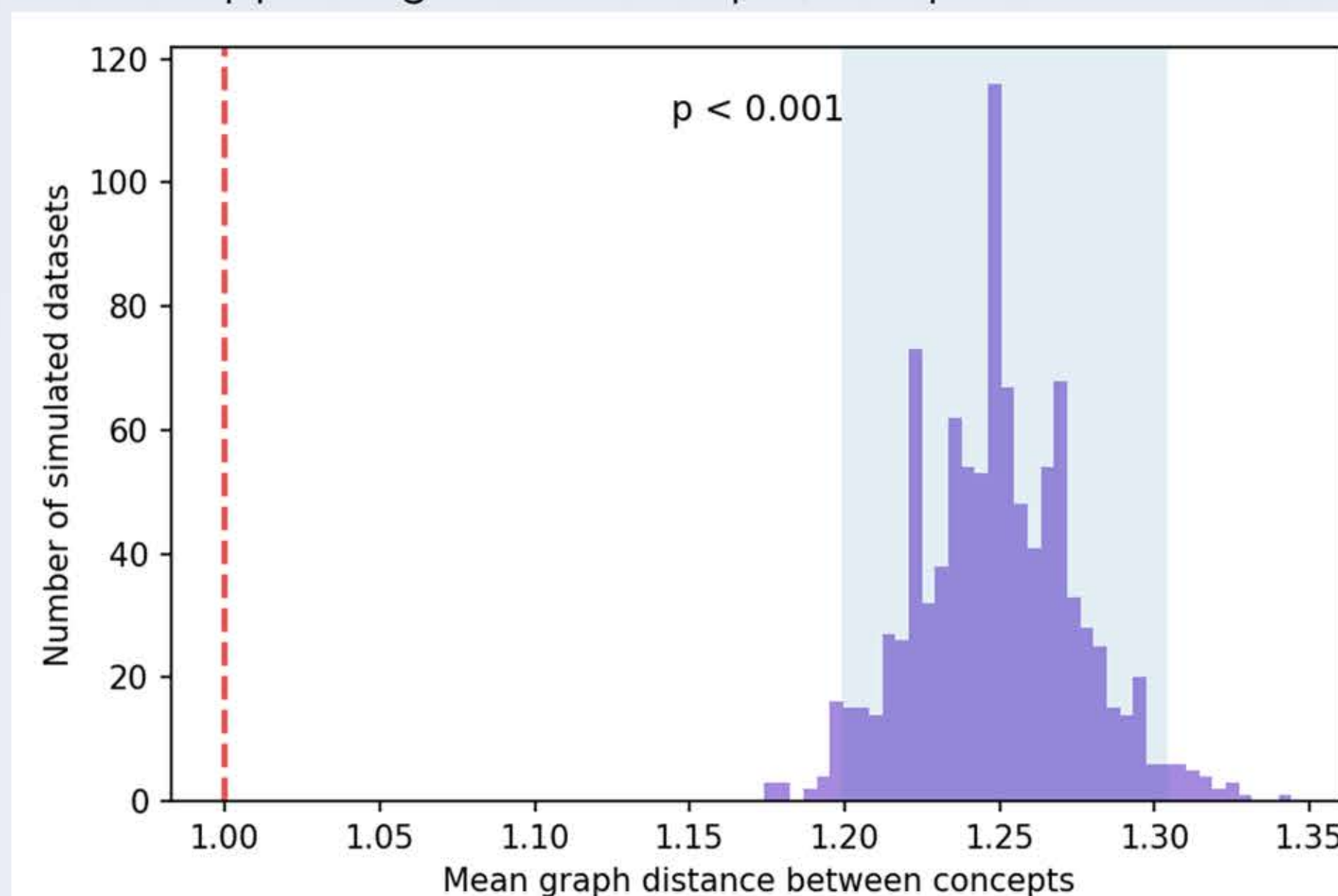
Children listed more features per concept on average compared to YA (filtering single occurrences).



Children produced more distinct features (i.e., features appearing in less concepts) compared to YA.



Children produced features more densely correlated (i.e., co-occurring more frequently) compared to YA.



Children produced features such that all pairs of concepts had at least one co-occurring feature in them.

Figure 3–6. We compared a subset ($n = 25$ concepts) of the child dataset to a subset of the YA benchmark with overlapping concepts [8, 9]. Since the YA benchmark contained $n = 20$ responders to concepts (compared to our $n = 15$), we ran an 1000-fold bootstrap downsampling the YA benchmark to $n = 15$ using binominal distribution. Purple histograms show the mean of simulated datasets on specific variables, while the red line and area represents the mean and standard error of sample.

Acknowledgments

This work was supported by the Hungarian National Research, Development and Innovation Office – NKFIH (FK 146496), a Bolyai János Research Scholarship of the Hungarian Academy of Sciences, and a Max Planck Partner Group by the Max Planck Society awarded to Attila Keresztes.

References

- McRae K, Cree GS, Seidenberg MS, McNorgan C. Semantic feature production norms for a large set of living and nonliving things. Behav Res Methods. 2005; november 1;37(4):547–59.
- Hovhannisyan M, Clarke A, Geib BR, Cicchinelli R, Monge Z, Worth T, és mtsai. The visual and semantic features that predict object memory: Concept property norms for 1,000 object images. Mem Cognit. 2021; május 1;49(4):712–31.
- Trón V, Kornai A, Gyepesi G, Németh L, Halácsy P, Varga D. Hunmorph: open source word analysis. In: Proceedings of the Workshop on Software. USA: Association for Computational Linguistics; 2005. o. 77–85. (Software '05).
- Indig B, Sass B, Simon E, Mittelholcz I, Vadász N, Makrai M. One format to rule them all – The emtsv pipeline for Hungarian. In: Friedrich A, Zeyrek D, Hoek J, szerkesztő. Proceedings of the 13th Linguistic Annotation Workshop [Internet]. Retrieved: <https://aclanthology.org/W19-4018/>

- McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction [Internet]. arXiv; 2020.
- Gholizadeh N, Saadatfar H, Hanaifi N. K-DBSCAN: An improved DBSCAN algorithm for big data. J Supercomput. 2020; november 26;77(6):6214–35.
- Heredia C, Moreno S, Canessa E, Chaigneau S. WordListsAnalytics: An R package for multiple data analysis of Property Listing Tasks. SoftwareX. 2024; szeptember 1;27:101842.
- Vinson, D. P., & Vigliocco, G. (2008a). Semantic feature production norms for a large set of objects and events. Behavior Research Methods, 40(1), 183–190. <https://doi.org/10.3758/BRM.40.1.183>
- Vinson, D. P., & Vigliocco, G. (2008b). Vinson-BRM-2008a.zip. Retrieved April 17, 2025 from Springer Nature <https://static-content.springer.com/esm/art%3A10.3758%2FBRM.40.1.183/MediaObjects/Vinson-BRM-2008a.zip>

Take-away

Acquiring concept features from children aged 6–9 years is feasible on a large scale.

Children generated a surprisingly rich set of features, yielding a well-clustered yet overlapping network across canonical categories. The resulting dense but expansive network suggests an uneven development of vocabulary and knowledge organization, with many words mapping onto the same meanings.

We plotted two UMAP-based [5] principal components and clusters via DBSCAN [6]

